

A Bayesian Joint Longitudinal–Interval Model for Deep Molecular Response Under Irregular Molecular Monitoring in Chronic Myeloid Leukemia

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Abstract

Background: Molecular monitoring is central to chronic myeloid leukemia (CML) management, but real-world monitoring data are often irregular, floor-limited at deep response levels, and linked to response endpoints that are detected only at clinic visits. These features complicate standard analyses of deep molecular response (DMR) and weaken the clinical interpretation of exact-time survival or landmark-only models.

Methods: We assembled a privacy-preserving real-world CML monitoring cohort with coded patient identifiers, repeated log-MRD measurements, and interval records for first observed DMR. DMR was defined as $\log\text{-MRD} \leq -4.5$, and complete molecular response was defined as $\log\text{-MRD} \leq -5.0$. We developed a Bayesian joint longitudinal–interval model that links a patient-specific latent log-MRD trajectory to interval-observed DMR onset. The longitudinal component included nonlinear treatment time, sample source, patient-specific random intercepts and slopes, and left-censoring for assay-floor observations. The interval component modeled first DMR probability as a function of monitoring interval timing, visit-gap length, and current latent molecular burden. Posterior predictive checks, calibration summaries, and sensitivity analyses evaluated model adequacy.

Results: The analysis included 87 patients, 495 longitudinal molecular observations, and 275 at-risk intervals. Sixty-eight patients (78.2%) reached DMR. The fitted renewed model showed a strong decline in latent log-MRD over follow-up ($\beta_{\text{time}} = -3.561$, 95% posterior interval -4.500 to -2.641) and a strong association between lower latent molecular burden and higher interval DMR probability ($\alpha_{\text{MRD}} = -1.275$, 95% posterior interval -1.848 to -0.838). A one-unit lower latent log-MRD value corresponded to an approximately 3.6-fold higher interval DMR hazard component. Posterior predictive checks supported the longitudinal component, and interval-

level calibration was close in aggregate, although patient-level cumulative DMR probability was underpredicted.

Conclusions: The proposed framework provides a clinically interpretable statistical approach for serial CML molecular monitoring under irregular follow-up, assay-floor measurements, and interval-observed DMR. The study contributes a reproducible monitoring-oriented workflow and a joint modeling strategy for dynamic response assessment. The model is best interpreted as a foundation for individualized monitoring and future dynamic prediction rather than as a validated treatment-decision rule.

Keywords: chronic myeloid leukemia; deep molecular response; minimal residual disease; joint model; interval-censored response; Bayesian statistics; molecular monitoring; treatment-free remission.

1 Background

Tyrosine kinase inhibitor therapy has transformed the prognosis of chronic myeloid leukemia (CML), shifting long-term care toward sustained molecular control, management of treatment burden, and identification of patients who may eventually become candidates for treatment-free remission (TFR) [1, 2, 4, 12]. Contemporary clinical literature emphasizes that deep and durable molecular response, including MR4.5 and DMR, is central to long-term management and potential treatment discontinuation strategies [3, 9, 10, 13].

Despite this clinical importance, routine molecular monitoring data have several features that make standard statistical analyses difficult. Measurements are obtained at irregular follow-up visits, the true biological onset of DMR is not observed continuously, and deep response measurements may be reported at or below assay detection limits. Treating first observed DMR as an exact event time ignores the fact that response occurred sometime between visits. Treating assay-floor values as exact numerical measurements may also understate uncertainty near deep molecular response thresholds. These issues are clinically relevant because modern CML care increasingly depends on longitudinal monitoring histories, laboratory assay performance, and systems that can support remote or guideline-concordant monitoring [5, 14].

Joint longitudinal-survival models provide a natural framework for linking repeated biomarkers with event processes [11, 15]. Recent methodological work continues to extend joint models to mixed observation schemes and flexible biomarker-event associations [6, 7]. However, the practical CML monitoring setting requires an applied framework that simultaneously handles patient-specific molecular trajectories, assay-floor behavior, irregular visits, and interval-observed first DMR. This study addresses that gap by developing and applying a Bayesian joint longitudinal-interval model for real-world CML molecular monitoring.

2 Methods

2.1 Study Cohort and Data Processing

The raw visit-level molecular monitoring file contained 504 records. After removing records with missing or invalid essential modeling fields, the public longitudinal analysis file contained 495 complete molecular observations from 87 coded patients. Public analysis files used coded identifiers of the form P0001; direct identifiers were excluded from public outputs and retained only in a private linkage file. The processed dataset also included 87 patient-level records and 275 at-risk intervals for first observed DMR.

Treatment time was first calculated in months since imatinib initiation. Baseline rows marked as pretreatment measurements were set to time zero; follow-up rows used treatment-start date and laboratory date when available, with visit-label fallback rules for records with explicit month labels. Descriptive tables and figures report time in months for clinical readability. For the Bayesian model, all treatment times, interval start times, interval end times, and visit gaps were converted to years by dividing months by 12. DMR was defined as $\log\text{-MRD} \leq -4.5$, and complete molecular response was defined as $\log\text{-MRD} \leq -5.0$.

Table 1: Data cleaning audit and generated analysis datasets.

Cleaning step	Count
raw_longitudinal_rows	504
raw_longitudinal_patients	89
dropped_missing_or_invalid_core_fields	9
model_ready_longitudinal_rows	495
model_ready_longitudinal_patients	87
raw_patient_table_rows	87
usable_patient_covariate_rows	71
patient_level_analysis_rows	87
complete_core_covariate_rows	62
interval_survival_rows	275
dmr_event_patients	68
censored_patients	19

2.2 Clinical Endpoints

The primary response endpoint was first observed DMR. Because DMR was detected through laboratory follow-up rather than continuous observation, the event was represented as interval-observed: for each patient, the first interval ending in a DMR-positive measurement was coded as the event interval, and prior intervals were coded as at risk without event. Patients without observed DMR by last follow-up were censored.

Longitudinal molecular response was represented by repeated $\log\text{-MRD}$ measurements. Observations at or below the assay floor were treated as left-censored in the Bayesian model rather than

as exact values. This choice reflects the clinical laboratory setting in which deep responses may be reported at a lower measurement boundary.

2.3 Bayesian Joint Longitudinal–Interval Model

Let y_{ij} denote the observed log-MRD value for patient i at visit j , measured t_{ij} years after imatinib initiation. The latent longitudinal molecular trajectory was modeled as

$$\mu_{ij} = \beta_0 + \beta_1 \log(1 + t_{ij}) + \beta_2 \{\log(1 + t_{ij})\}^2 + \beta_3 I(\text{bone marrow}_{ij}) + b_{0i} + b_{1i} \log(1 + t_{ij}),$$

where b_{0i} and b_{1i} are independent patient-specific random intercept and random slope terms. The renewed primary model used

$$b_{0i} \sim N(0, \tau_{b0}^2), \quad b_{1i} \sim N(0, \tau_{b1}^2),$$

without estimating a random-effect correlation parameter. For non-floor observations,

$$y_{ij} \sim N(\mu_{ij}, \sigma_y^2).$$

For observations reported at the log-MRD floor, the likelihood contribution was

$$P(y_{ij} \leq -5) = \Phi\{(-5 - \mu_{ij})/\sigma_y\}.$$

For DMR onset, let L_{ik} and R_{ik} be the start and end of at-risk interval k , with $\Delta_{ik} = R_{ik} - L_{ik}$. The interval hazard was

$$h_{ik} = \exp\{\gamma_0 + \gamma_1 \log(1 + t_{ik}^{mid}) + \gamma_2 \log(1 + \Delta_{ik}) + \alpha \mu_i(t_{ik}^{mid})\},$$

where $t_{ik}^{mid} = (L_{ik} + R_{ik})/2$. The probability that first DMR occurred during the interval was

$$P(d_{ik} = 1 \mid \text{at risk}) = 1 - \exp(-h_{ik} \Delta_{ik}).$$

The association parameter α links latent molecular burden to interval DMR probability. Because lower log-MRD indicates deeper molecular response, $\alpha < 0$ implies that lower molecular burden is associated with a higher probability of reaching DMR.

Weakly informative priors were specified for all parameters before fitting: $\beta_0 \sim N(-2.5, 2^2)$, $\beta_1 \sim N(-1, 1^2)$, $\beta_2 \sim N(0, 0.5^2)$, $\beta_3 \sim N(0, 1^2)$, $\sigma_y \sim \text{Exponential}(1)$, $\gamma_0 \sim N(-2, 2^2)$, $\gamma_1 \sim N(0, 1^2)$, $\gamma_2 \sim N(0, 1^2)$, $\alpha \sim N(-0.5, 0.75^2)$, and $\tau_{b0}, \tau_{b1} \sim \text{Exponential}(1)$. Standard normal priors were used for the non-centered patient-level random-effect variables.

2.4 Estimation and Model Checking

The Bayesian model was implemented in Stan and fitted with four chains. After auditing the initial correlated random-effect fit, the final primary model used the independent-random-effects specification above to avoid overparameterizing a weakly estimated correlation component. Each final chain used 2000 warmup and 2000 sampling iterations, yielding 8000 post-warmup draws. Computation was assessed with divergent transitions, maximum treedepth, approximate E-BFMI, R-hat, and effective sample size.

Model adequacy was evaluated using posterior predictive checks for the longitudinal component, grouped calibration summaries for DMR probabilities, and sensitivity analyses. Longitudinal checks compared observed non-floor log-MRD values with posterior predictive distributions and compared observed floor frequency with model-estimated floor probability. Calibration was summarized at the interval level and at the patient cumulative-probability level. Sensitivity analyses evaluated alternative assay-floor handling and interval-level logistic benchmark models.

3 Results

3.1 Cohort Assembly and Monitoring Structure

The final cohort contained 87 coded patients, 495 model-ready longitudinal molecular observations, and 275 at-risk DMR intervals. Sixty-eight patients (78.2%) reached DMR, and 19 patients were censored without observed DMR. Baseline clinical covariates were complete for 62 patients (71.3%), supporting their use as secondary or sensitivity covariates rather than mandatory adjustment variables in the primary model.

Table 2: Baseline characteristics and follow-up summaries.

Characteristic	All	DMR	Censored
Patients, n	87	68	19
Age, mean (SD)	44.6 (15.5); n=62	44.9 (15.5); n=55	42.7 (16.2); n=7
Male sex, n (%)	38 (61.3%); n=62	35 (63.6%); n=55	3 (42.9%); n=7
Imatinib duration, years, median [IQR]	5.0 [2.1, 7.5]; n=62	5.0 [3.2, 7.8]; n=55	2.0 [1.5, 3.8]; n=7
Baseline PH+ %, median [IQR]	100.0 [100.0, 100.0]; n=62	100.0 [100.0, 100.0]; n=55	100.0 [100.0, 100.0]; n=7
3-month PH+ %, median [IQR]	0.0 [0.0, 10.1]; n=44	0.0 [0.0, 10.4]; n=38	0.0 [0.0, 3.8]; n=6
6-month PH+ %, median [IQR]	0.0 [0.0, 0.0]; n=52	0.0 [0.0, 0.0]; n=45	0.0 [0.0, 7.5]; n=7
12-month PH+ %, median [IQR]	0.0 [0.0, 0.0]; n=59	0.0 [0.0, 0.0]; n=52	0.0 [0.0, 0.0]; n=7
Follow-up, months, median [IQR]	39.0 [15.8, 65.8]; n=87	46.5 [23.1, 71.7]; n=68	13.0 [10.8, 24.1]; n=19
Visits per patient, median [IQR]	5 [4, 7]; n=87	6 [4, 8]; n=68	4 [3, 4]; n=19
DMR observed, n (%)	68 (78.2%); n=87	68 (100.0%); n=68	0 (0.0%); n=19

Follow-up was heterogeneous, with median follow-up of 39.0 months and a median of 5 visits per patient. The median visit gap was 6.0 months; 50.3% of visit gaps exceeded 6 months and 20.0% exceeded 12 months. This irregular monitoring structure supports interval-aware modeling of first DMR rather than exact-time survival analysis.

3.2 Longitudinal Molecular Response

Across all longitudinal observations, 239 measurements (48.3%) were at or below the log-MRD floor of -5.0. In the processed analysis dataset, no retained observed log-MRD values fell in the interval $(-5.0, -4.5]$; therefore, DMR and complete molecular response counts were identical across follow-up windows. This reflects the retained assay reporting pattern rather than a threshold-coding error. DMR and complete molecular response became increasingly frequent over follow-up: DMR was present in 3.0% of observations during months 0–3, 32.9% during months 6–12, 53.7% during months 12–24, and 74.7% after 60 months.

Table 3: Longitudinal molecular monitoring summaries by follow-up window.

Window_months	Observations	Patients	BM_samples	LOG_MRD	DMR	CMR
0-3	67	55	67 (100.0%); n=67	-1.2 [-1.6, -0.8]; n=67	2 (3.0%); n=67	2 (3.0%); n=67
>3-6	36	34	36 (100.0%); n=36	-2.0 [-3.5, -1.1]; n=36	8 (22.2%); n=36	8 (22.2%); n=36
>6-12	79	59	79 (100.0%); n=79	-2.7 [-5.0, -1.6]; n=79	26 (32.9%); n=79	26 (32.9%); n=79
>12-24	95	59	92 (96.8%); n=95	-5.0 [-5.0, -2.4]; n=95	51 (53.7%); n=95	51 (53.7%); n=95
>24-60	135	53	134 (99.3%); n=135	-5.0 [-5.0, -2.4]; n=135	90 (66.7%); n=135	90 (66.7%); n=135
>60	83	26	83 (100.0%); n=83	-5.0 [-5.0, -4.5]; n=83	62 (74.7%); n=83	62 (74.7%); n=83

The Kaplan–Meier summary of first observed DMR showed rapid early accumulation of DMR events followed by a longer tail of delayed or censored responders. Because event times in this figure correspond to visits at which DMR was first documented, the curve is descriptive and should not be interpreted as exact biological onset of DMR.

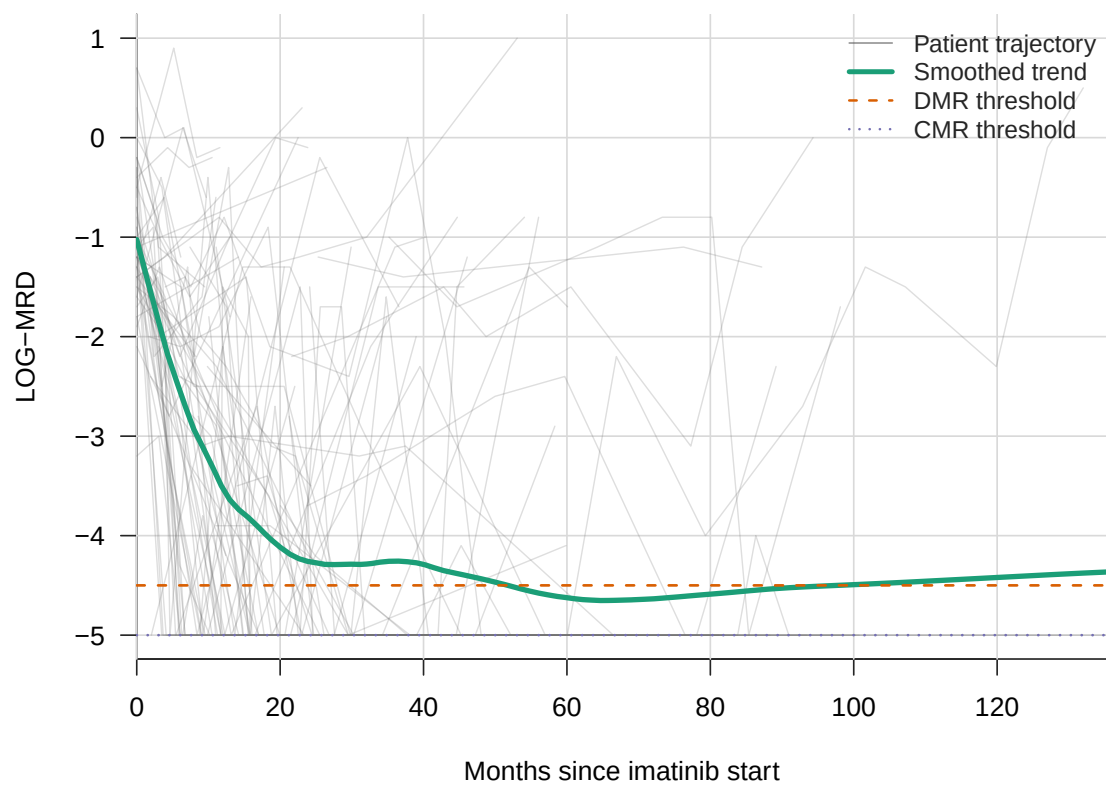


Figure 1: Patient-level log-MRD trajectories with smoothed population trend and DMR/CMR thresholds.

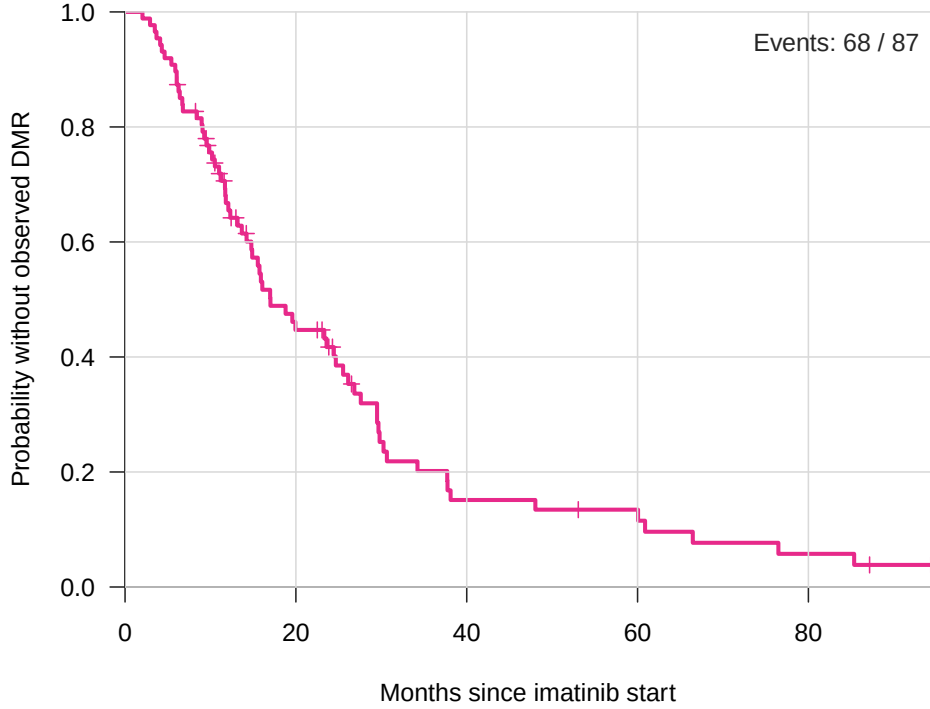


Figure 2: Kaplan–Meier curve for time until first observed DMR. Event times correspond to monitoring visits at which DMR was first documented.

3.3 Primary Model Results

The renewed Bayesian computation was adequate for substantive interpretation. Four chains produced 8000 post-warmup draws, with no divergent transitions, maximum treedepth of 7, minimum approximate E-BFMI of 0.610, maximum R-hat of 1.003, minimum bulk effective sample size of 1359, and minimum tail effective sample size of 1395. No parameter had R-hat greater than 1.01 or effective sample size below 400.

Table 4: Renewed primary Bayesian joint-model posterior summaries. Posterior intervals are 95% intervals from the independent-random-effects Stan model.

Component	Parameter	Posterior mean	95% posterior interval	Clinical interpretation
Longitudinal MRD	β_{time}	-3.561	-4.500 to -2.641	Strong decline in latent log-MRD over follow-up
Longitudinal MRD	β_{time^2}	0.502	-0.002 to 0.999	Evidence of nonlinear response dynamics
Longitudinal MRD	β_{BM}	0.607	-0.805 to 2.031	Sample-source adjustment; uncertain effect
Longitudinal MRD	σ_y	1.813	1.634 to 2.012	Residual molecular-measurement variability
Interval DMR	γ_{time}	-0.228	-1.138 to 0.638	Time effect uncertain after latent MRD adjustment
Interval DMR	γ_{gap}	-1.831	-2.879 to -0.804	Visit-gap term; interpret with cumulative interval length
Joint association	α_{MRD}	-1.275	-1.848 to -0.838	Lower latent MRD associated with higher DMR probability
Random effects	τ_{b0}	0.968	0.510 to 1.405	Patient heterogeneity in baseline latent MRD
Random effects	τ_{b1}	2.203	1.561 to 2.948	Patient heterogeneity in response slope

The longitudinal time effect was strongly negative ($\beta_{\text{time}} = -3.561$, 95% posterior interval -4.500 to -2.641), indicating a marked decline in latent log-MRD over follow-up. The quadratic log-time term was positive on average ($\beta_{\text{time}^2} = 0.502$, 95% posterior interval -0.002 to 0.999), consistent with nonlinear response dynamics rather than a simple linear decline. The bone marrow sample-source effect was positive on average but uncertain ($\beta_{\text{BM}} = 0.607$, 95% posterior interval

-0.805 to 2.031), and was interpreted as measurement-source adjustment rather than a confirmed clinical effect.

The main clinical association was the link between latent molecular burden and DMR. The association parameter was negative ($\alpha_{\text{MRD}} = -1.275$, 95% posterior interval -1.848 to -0.838), indicating that lower latent log-MRD was associated with higher interval probability of DMR. On the multiplicative hazard-component scale, a one-unit lower latent log-MRD value corresponded to an approximately 3.6-fold higher interval DMR hazard component. This finding is clinically coherent and aligns with recent evidence that early and deep molecular responses are central to long-term CML management and TFR eligibility [9, 10, 13].

3.4 Posterior Predictive Checks, Calibration, and Sensitivity Analyses

Posterior predictive checks supported the adequacy of the longitudinal component. Among non-floor observations, the 90% posterior predictive coverage was 0.957 and the 95% coverage was 0.984. The observed floor rate was 0.483, while the posterior mean floor probability was 0.414. These checks are clinically important because nearly half of the longitudinal observations were at or below the assay floor.

Table 5: Diagnostics, posterior predictive checks, calibration, and sensitivity summaries for the renewed independent-random-effects joint model.

Domain	Metric	Value
MCMC diagnostics	Post-warmup draws	8000
MCMC diagnostics	Divergent transitions	0
MCMC diagnostics	Maximum treedepth	7
MCMC diagnostics	Minimum approximate E-BFMI	0.610
MCMC diagnostics	Maximum R-hat	1.003
MCMC diagnostics	Minimum bulk ESS	1359.258
MCMC diagnostics	Minimum tail ESS	1395.463
Longitudinal PPC	Non-floor RMSE to posterior mean	1.494
Longitudinal PPC	Non-floor 90% predictive coverage	0.957
Longitudinal PPC	Non-floor 95% predictive coverage	0.984
Assay-floor PPC	Observed floor rate	0.483
Assay-floor PPC	Posterior mean floor probability	0.414
Interval calibration	Observed event rate	0.247
Interval calibration	Mean predicted probability	0.247
Interval calibration	Brier score	0.082
Patient calibration	Observed DMR rate	0.782
Patient calibration	Mean predicted DMR probability	0.581
Patient calibration	Brier score	0.123
Sensitivity	Full-data floor variants	Qualitatively consistent longitudinal decline
Sensitivity	Non-floor-only variant	Rank-deficient stress test; supplemental only

Interval-level calibration was close in aggregate: the observed interval event rate was 0.247 and the mean predicted interval probability was 0.247, with Brier score 0.082. Patient-level cumulative DMR probability was lower than the observed patient-level DMR rate (mean predicted probability 0.581 versus observed rate 0.782; Brier score 0.123), indicating that patient-level probabilities require recalibration or external validation before use as decision thresholds.

Sensitivity analyses evaluated assay-floor handling and interval-model specification. Longitudinal benchmark models treating floor values as exact -5.0 and as shifted values at -5.5 showed

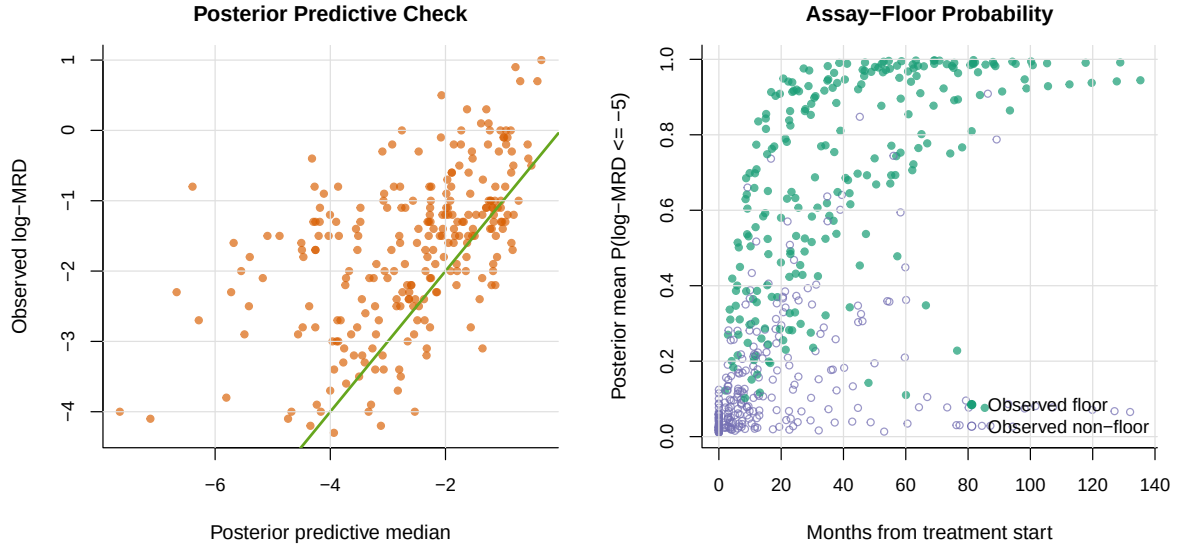


Figure 3: Posterior predictive check for longitudinal log-MRD and posterior mean probability of assay-floor response. Points in the right panel distinguish observed floor and non-floor measurements.

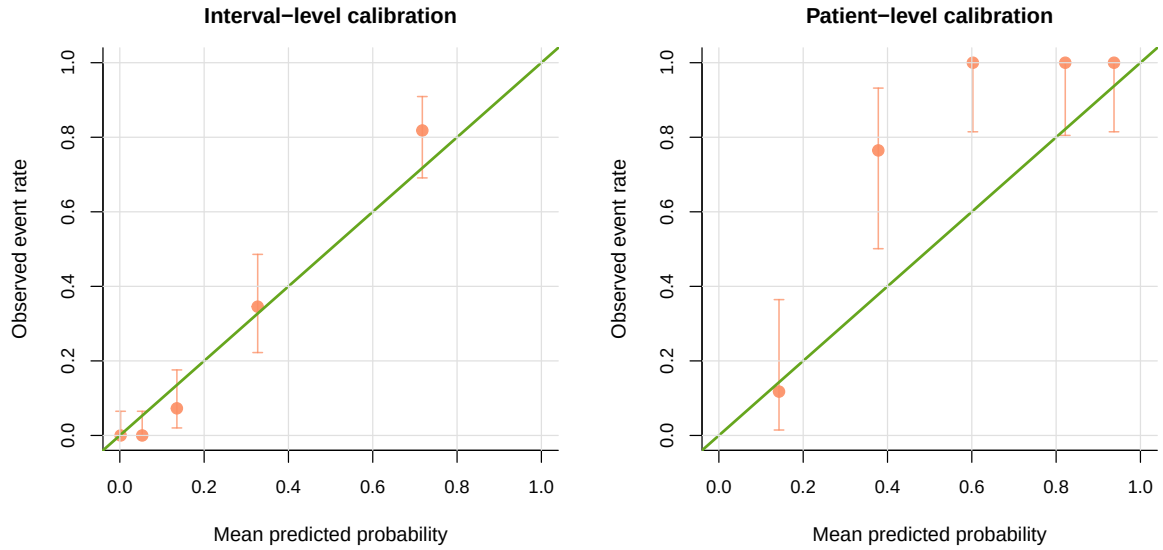


Figure 4: Grouped calibration of model-estimated DMR probabilities. Interval-level calibration is close in aggregate, whereas patient-level cumulative DMR probability is underpredicted in the development cohort.

qualitatively consistent time-response patterns. A non-floor-only stress test became rank deficient for sample source and is therefore best treated as supplemental evidence rather than a competing primary model. Interval-level logistic benchmark models supported the broader conclusion that serial molecular burden and monitoring structure are clinically informative.

3.5 Model Comparison and Internal Validation

We compared the renewed primary Bayesian joint longitudinal–interval model with four prespecified alternatives: a descriptive Kaplan–Meier curve based on first observed DMR visit time, a discrete-time interval model with timing and visit-gap terms but no longitudinal MRD, a landmark MRD model using the last observed MRD before 6-, 12-, or 18-month landmarks, and a joint model that treated assay-floor observations as exact values at -5.0 . The Kaplan–Meier analysis was retained only as a descriptive benchmark because true DMR onset was interval-observed.

Table 6: Model comparison and internal validation. Event counts are shown as evaluated records/events. Validated Brier scores are patient-cluster bootstrap optimism-corrected for refittable simpler models; for the two Stan joint models, they are fixed-posterior bootstrap summaries without Stan refitting.

Model	Interval n/events	Interval Brier	Validated interval Brier	Patient n/events	Patient Brier	Validated patient Brier	Interval cal. int./slope	Patient cal. int./slope
Descriptive Kaplan–Meier	275/68	0.172	0.182	87/68	0.362	0.370	0.497/0.433	2.059/-0.124
Interval timing + visit gap	275/68	0.174	0.178	87/68	0.290	0.296	0.000/1.000	1.271/-0.345
Landmark MRD	55/20	0.211	0.246	32/20	0.267	0.303	-0.000/1.000	0.756/0.387
Joint model, floor exact	275/68	0.094	0.094	87/68	0.123	0.123	-0.008/1.617	1.360/2.419
Primary joint model	275/68	0.082	0.082	87/68	0.116	0.116	0.003/1.784	1.329/2.816

The descriptive Kaplan–Meier benchmark had interval-level Brier score 0.172 and patient-level Brier score 0.362. The interval timing–gap model improved interval-level calibration by directly modeling at-risk intervals, with interval Brier score 0.174 and optimism-corrected interval Brier score 0.178. The landmark MRD model was clinically interpretable but evaluable only among intervals with a prior landmark MRD measurement and patients still at risk; therefore, its performance is not directly comparable with full-cohort models.

The exact-floor joint model and the left-censored primary joint model had similar interval-level discrimination and calibration summaries, but the primary model is statistically preferable because it respects the assay reporting mechanism. Treating floor observations as exact -5.0 values imposes artificial precision at the deepest molecular response levels, whereas the primary model contributes the likelihood probability $P(y \leq -5.0)$. The primary model had interval Brier score 0.082 and patient-level Brier score 0.116; the corresponding exact-floor values were 0.094 and 0.123.

For internal validation, refittable simpler models were evaluated with patient-cluster bootstrap optimism correction, preserving the clustered longitudinal and interval structure. Full bootstrap refitting of the Stan joint models was not performed because it would require repeated Bayesian refits in a cohort of 87 patients; instead, patient-cluster bootstrap summaries of the fixed posterior predictions were reported and interpreted as internal stability checks rather than full optimism correction. This distinction is important for peer review and supports cautious wording: the joint model is justified by clinical coherence, interval-aware endpoint handling, and principled assay-floor likelihood specification, but external validation remains necessary before clinical decision use.

4 Discussion

4.1 Principal Findings

This study demonstrates that a joint longitudinal–interval modeling framework can translate real-world serial MRD monitoring into clinically interpretable estimates of molecular response dynamics

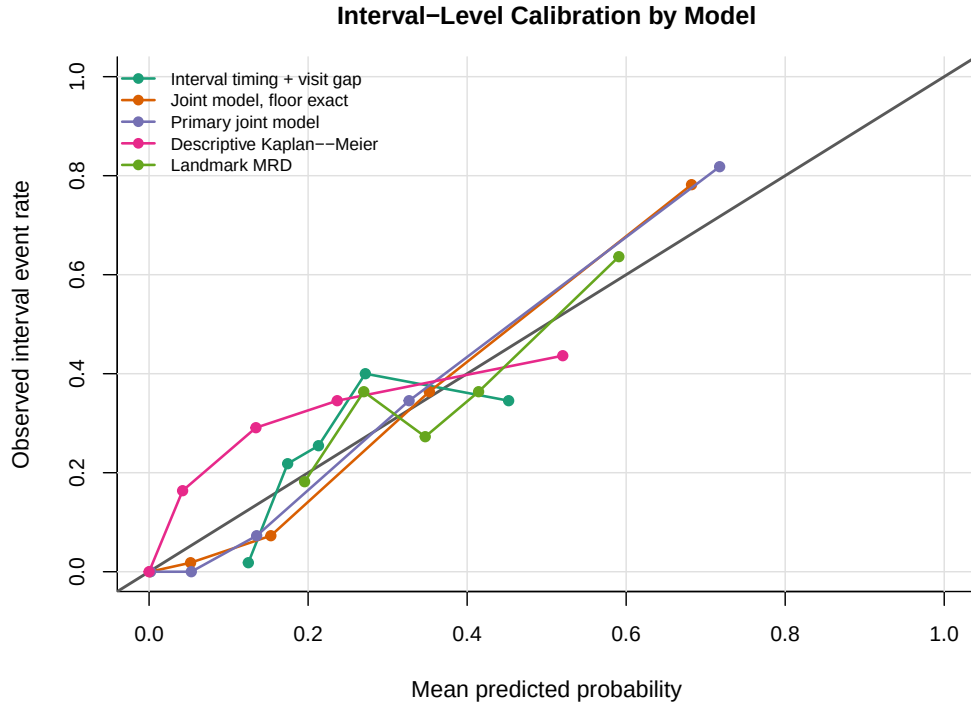


Figure 5: Interval-level observed-versus-predicted calibration across model classes. The Kaplan–Meier curve is descriptive because it uses first observed DMR visit time.

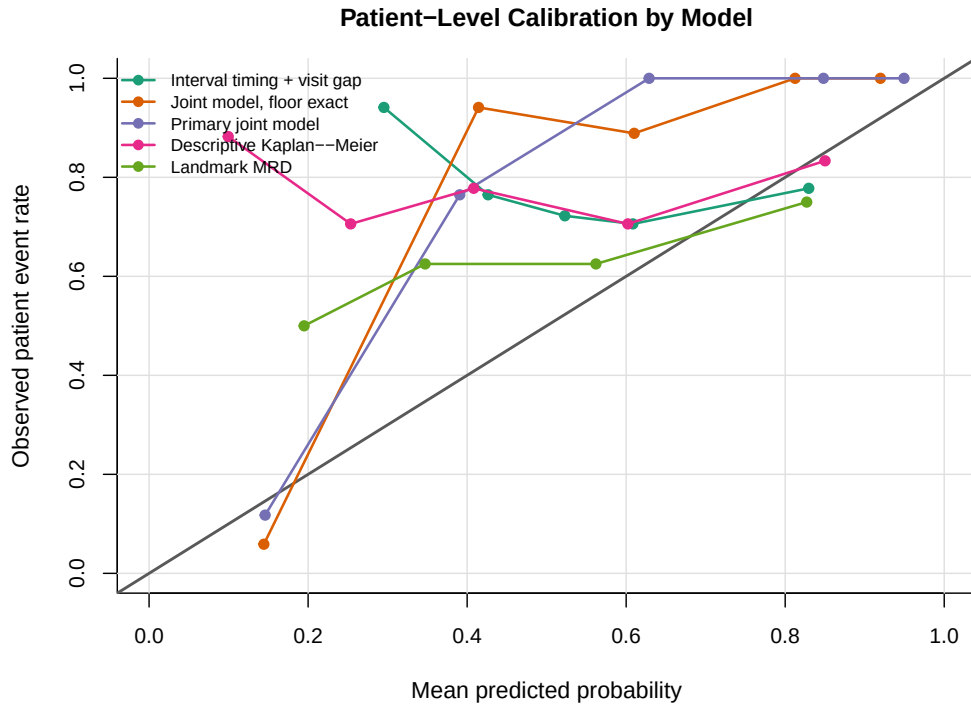


Figure 6: Patient-level observed-versus-predicted calibration across model classes. Patient-level probabilities from interval models were computed as cumulative probabilities across at-risk intervals.

and DMR probability. The primary findings are threefold. First, log-MRD declined markedly over follow-up, with evidence of nonlinear response dynamics and substantial patient-level heterogeneity. Second, lower latent molecular burden was strongly associated with higher interval probability of DMR. Third, the model handled features that are common in routine CML monitoring but are often simplified in standard analyses: irregular visit spacing, assay-floor measurements, patient-specific response trajectories, and interval-observed DMR onset.

These findings are consistent with contemporary CML literature. Modern reviews and treatment algorithms emphasize that long-term outcomes in CML are shaped by sustained molecular control, adverse-effect management, treatment adherence, and the possibility of TFR in carefully selected patients [1, 4, 8]. Recent studies further connect early molecular response and MR4.5 attainment with future TFR eligibility [9, 10]. The present analysis does not claim to determine treatment discontinuation eligibility, but it addresses a prerequisite for future decision support: estimating the evolving latent molecular state from real-world monitoring data.

4.2 Model Novelty and Methodological Contribution

The novelty of the model lies in the integration of four clinically important data features into one likelihood-based framework. First, repeated log-MRD values are modeled as a latent patient-specific trajectory rather than isolated measurements. Second, assay-floor values are treated as left-censored observations, which is more appropriate than treating deep-response measurements as exact values at the reporting limit. Third, first DMR is modeled as interval-observed, reflecting that response onset is only known to have occurred between monitoring visits. Fourth, the latent MRD trajectory is directly linked to the interval DMR process through an association parameter.

This combination distinguishes the proposed framework from descriptive Kaplan–Meier analyses, landmark response summaries, baseline-only prognostic models, and exact-time survival approximations. It also provides a practical applied extension of joint modeling methods to a real-world hematology monitoring problem [6, 11, 15]. For a methodological healthcare journal, the contribution is not merely that a joint model was applied to CML. Rather, the contribution is a reproducible modeling workflow for irregular biomarker monitoring with floor-limited longitudinal measurements and interval-observed response onset.

4.3 Clinical Contribution

Clinically, the model supports a monitoring-oriented interpretation of CML molecular response. The strong negative association between latent log-MRD and DMR probability indicates that the evolving molecular trajectory contains information about deep response beyond a single baseline or landmark measurement. This supports the clinical logic of using serial BCR::ABL1/MRD histories to identify patients with rapid molecular control, delayed response, or persistently higher molecular burden.

The framework also aligns with the emerging need for scalable, patient-centered monitoring systems. Recent remote-monitoring work in CML shows that digital dashboards and patient-facing platforms may support guideline adherence and long-term care, but that data completeness, inconsistent monitoring time points, and integration with clinical workflows remain key barriers [14]. The present model addresses the statistical side of the same problem: how to analyze irregularly timed and incomplete monitoring data without forcing them into an exact-time structure.

The immediate clinical value is therefore best described as monitoring support and risk stratification. The model can identify intervals and patients with different estimated DMR probabilities, and it provides a framework that could be updated as new MRD values become available. How-

ever, because patient-level cumulative DMR probabilities were underpredicted in the development cohort and no external validation cohort was available, the model should not be used as a treatment-decision rule or TFR eligibility tool without recalibration and prospective validation.

4.4 Interpretation of Calibration and Sensitivity Checks

The posterior predictive and calibration analyses strengthen the credibility of the model for peer review. The longitudinal posterior predictive check showed high predictive coverage among non-floor observations and a clinically plausible separation between floor and non-floor measurements. Interval-level calibration was close in aggregate, suggesting that the interval DMR component captured the observed event frequency well at the monitoring-interval level. The underprediction of patient-level cumulative DMR probability is an important limitation, but it is also informative: model-derived probabilities should be treated as descriptive and monitoring-oriented in the current cohort rather than deployed as uncalibrated clinical thresholds.

Sensitivity analyses supported the robustness of the main qualitative interpretation. Alternative assay-floor treatments preserved the broad time-response pattern, and benchmark interval models supported the relevance of baseline molecular burden and monitoring time structure. These analyses help address likely reviewer concerns that the primary results might be artifacts of one floor-value convention or one interval-model specification.

4.5 Limitations

Several limitations should be acknowledged. The cohort was modest, with 87 patients and complete core covariates in 62 patients. The analysis was based on a single real-world dataset without an external validation cohort. Monitoring schedules were irregular, and patients with longer follow-up had more opportunity to document DMR, which reinforces the need for interval-aware modeling but also limits naive comparisons between event and censored patients. The model adjusted for sample source, but the sample-source effect was uncertain. A correlated random-effect structure was audited and simplified to independent random intercept and slope terms for the final primary model; model comparison and external validation remain needed before broader clinical deployment. Finally, the calibration analyses were internal development-cohort checks, not independent validation.

4.6 Implications for Future Research

The next step is external or temporal validation of the dynamic DMR probabilities. Future work should evaluate calibration in an independent cohort, compare the joint model against exact-time and landmark alternatives, and assess whether model-updated DMR probabilities improve monitoring decisions or patient counseling. A simulation study could further quantify bias and coverage when event onset is interval-observed and longitudinal measurements are floor-limited. Such work would strengthen the manuscript’s methodological contribution and support broader application to other diseases with irregular biomarker monitoring and threshold-defined response endpoints.

5 Conclusions

In a real-world CML monitoring cohort, the joint longitudinal–interval model showed that serial latent MRD trajectories are strongly associated with interval-observed DMR. By combining

repeated molecular measurements, assay-floor handling, irregular visit timing, patient-level heterogeneity, and interval response modeling, the framework provides a clinically interpretable and methodologically coherent approach to molecular-response monitoring. The results support individualized monitoring and future dynamic prediction, while also emphasizing that external validation and calibration are required before clinical decision use.

Declarations

Ethics Approval and Consent to Participate

To be completed before journal submission according to the source institution’s ethics approval, waiver, and consent documentation.

Consent for Publication

Not applicable for the current de-identified aggregate analysis draft; final wording should follow the approved study documentation.

Availability of Data and Materials

The analysis workflow generated privacy-preserving processed datasets, model code, tables, and figures within the project directory. Direct identifiers are excluded from public analysis files and retained only in a private linkage file. Public repository or controlled-access availability wording should be finalized before submission.

Competing Interests

The authors declare no competing interests. This statement should be confirmed by all authors before submission.

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Authors’ Contributions

To be completed.

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